Practical:04	Implementation and Time analysis of factorial program using iterative and recursive method
__/___/2026	

Objective:

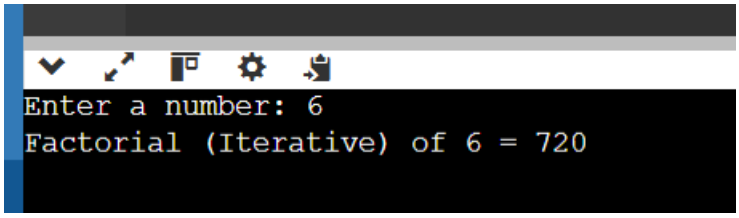
To implement factorial calculation using both iterative and recursive methods in C++, and to evaluate their execution time, memory utilization, and system-level overhead.

Program:

a) Iterative Method

```
#include <iostream>
using namespace std;
long factorialIterative(int n) {
    long result = 1;
    for (int i = 2; i <= n; i++) {
        result *= i;
    }
    return result;
}
int main() {
    int n;
    cout << "Enter a number: ";
    cin >> n;
    cout << "Factorial (Iterative) of " << n << " = "
         << factorialIterative(n) << endl;
    return 0;
}
```

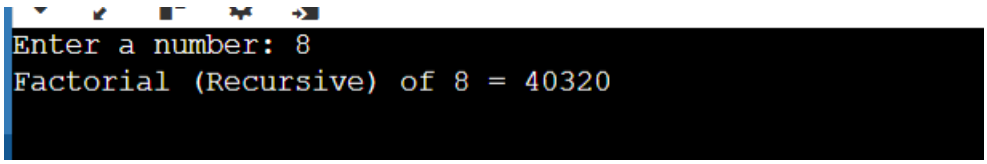
Output:



b) Recursive Method

```
#include <iostream>
using namespace std;
long factorialRecursive(int n) {
    if (n <= 1)         // base case
        return 1;
    return n * factorialRecursive(n - 1);  // recursive call
}
int main() {
    int n;
    cout << "Enter a number: ";
    cin >> n;
    cout << "Factorial (Recursive) of " << n << " = "
         << factorialRecursive(n) << endl;
    return 0;
}
```

Output:



Analysis:

Method	Time Complexity	Space Complexity	Stack Overhead
Iterative Method	$O(n)$	$O(1)$	**None**
Recursive Method	$O(n)$	$O(n)$	**High (proportional to n)**

Conclusion:

Practical testing shows that both iterative and recursive methods compute factorials quickly for small numbers, but iteration is safer and more memory-efficient because it avoids function call stack overhead and stack overflow risks.